

Their Wisdom via their Voices, Always with us.

Ask in your own language. Hear the voice you always wanted to hear.

CONNECTING OUR ANCESTORS TO THE GENERATIONS STILL TO COME

heritagevoice.org

PROBLEM AND SOLUTION

Spoken knowledge is fading. We keep it, with consent.

THE PROBLEM		HERITAGEVOICE AI
1 Recordings nobody can search	→	Transcribed, translated and searchable in your own language
2 AI that guesses and invents	→	Answers cite the exact recording and moment. Press play to hear it.
3 No control over who hears	→	Consent first. People approve. Families decide who hears.

Their voices, always with us.

ARCHITECTURE

One small system, built in clear layers

USE

- **Streamlit app** — Library, Ask, Connections, record, review
- **Amazon Cognito** — Sign-in on a separate secure domain

THINK

- **FastAPI + worker** — API and background jobs
- **LangGraph agents** — Seven agents, one job each, with fall-backs
- **Amazon Transcribe** — Timed transcripts from speech

REMEMBER

- **PostgreSQL** — The source of truth
- **Pinecone** — Meaning search, rebuildable
- **Neo4j** — Reviewed graph, rebuildable
- **Private S3** — Audio, encrypted

Runs on one small AWS instance. Built with Terraform. Deployed only by GitHub Actions, with no stored AWS keys.

Live now: heritagevoice.org · demo library in 9 languages

HOW THE AGENTS WORK TOGETHER

One question, followed step by step

A FAMILY MEMBER ASKS, IN ENGLISH

"What does it teach?" (right after reading the Tamil proverb about humility)

1	**Question rewriter**	Makes it stand alone: "What does the Tamil proverb about humility teach?"
2	**Supervisor**	Chooses the route: this is about a passage and a relationship, so use both searches.
3	**Hybrid RAG + GraphRAG**	Not agents, they find evidence: the passage where it is told, and the reviewed link "proverb teaches humility". Duplicates are merged.
4	**Verifier**	Checks each claim against the evidence and drops anything the recordings do not support.
5	**Answer writer**	Writes the answer from the verified claims only. Code, not the AI, adds the citations.

Answer: "It teaches humility." with a link to the recording. When a recording is added, 3 more agents help earlier: translator, graph extractor and correction interpreter.

HYBRID RAG, AS BUILT

Two ways to search, fused into one list

1 · WHEN A RECORDING IS PUBLISHED

Approved text
Transcript and English translation

→

Chunking
By **speaker turn**, 100–250 tokens. A segment is never split, so every chunk keeps its time in the audio.

→

PostgreSQL
Chunk text, for keyword search

Pinecone
1,024-number vectors, no text. One namespace per community.

←

Embedding model
multilingual-e5-large, cosine similarity

2 · WHEN SOMEONE ASKS A QUESTION

Question
Already rewritten to stand alone

→

Keyword search
PostgreSQL full text. Exact names and Tamil words. Top 20.

Meaning search
Question embedded, nearest vectors in Pinecone. Top 20.

→

Permission check
Every hit reloaded from PostgreSQL. Not approved, consented and readable by you: dropped.

→

Rank fusion
Reciprocal rank fusion merges both lists and removes duplicates.

→

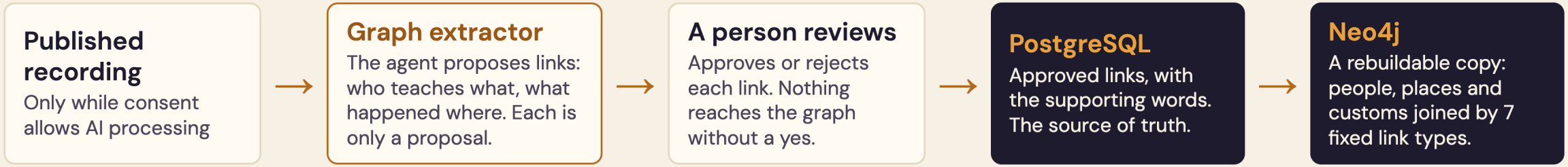
Best 5–8
At most 2 per recording. On to the verifier.

If Pinecone is down, keyword search carries on and the answer shows a warning. Pinecone can always be rebuilt from PostgreSQL.

GRAPHRAG, AS BUILT

Only reviewed relationships, double-checked on every question

1 · BEFORE THE GRAPH: PROPOSED BY AI, APPROVED BY PEOPLE



2 · WHEN SOMEONE ASKS A QUESTION



If Neo4j is not connected or is down, the question falls back to text search and the answer shows a warning. The graph can never show more than the text could.

OUTCOME

It works end to end, and we measured it.

24/24
test questions found the right recording in the top 5

9
languages in the live demo library, all searchable

130ms
typical search time, including the embedding call

1,400+
automated tests guard privacy and behaviour

Honest note: the test set is small and synthetic. With a simple word-matching index we found 23 of 24; real multilingual embeddings found all 24, including the Tamil paraphrase.